Graduation Disparities: Exploring Minority Group in Higher Education

Zaray Aguilar

University of California, San Diego

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Abstract

This report is meant to analyze the relationship between minority students and graduation rates throughout various universities in the United States. By focusing on Asian/Pacific Islander, Black, Hispanic, and Non-resident Alien student populations, the data demonstrates how enrollment size and average SAT scores can contribute to the disparities in graduation rates among minority students. When looking at the research question, *How does the relationship between minority dynamics and graduation rates vary among different minority groups?* The data seems to support the hypothesis that graduation rates for minority students will decline throughout universities across the U.S. This may be due to differences in population size, minority dynamics and graduation rates, but not all minority groups are impacted in the same way.

Introduction:

Higher education is presented as a gateway towards opportunities where students believe that a degree will create a path towards a successful future. Though this idea of higher education may be true for many students that attend colleges and universities throughout the United States, the reality is that not all students are given the same opportunities, resources, and experiences. That is to say, even though higher education is a critical aspect of being able to obtain a successful career there are still significant facts that have the potential to impact the experiences of students. This report aims to address how minority students may show a decrease in graduation percentage in the year 2000. The main question that is being addressed in this paper is *How does the relationship between minority students and graduation rates vary among different minority groups?* This question takes into account how educational environments can contribute to academic success and graduation rates. When taking into account factors such as SAT scores and graduation rates, there is a correlation that shows how scores may reflect not just how students perform academically but also how there may be systemic factors that can affect a students' educational experience. Students may have varying graduation rates that can be attributed to unequal opportunities to access resources compared to other students. By focusing on groups of students including Asian/Pacific Islander, Black, Hispanic, and Non-resident Alien student populations, the data demonstrates how enrollment size and average SAT scores can contribute to the differences in graduation rates among minority students. Potential confounding variables in college graduation rates can consider socioeconomic factors or educational support.

The data that was collected for the research comes from Opportunity Insights and focuses on data from Universities across the United States from 2000 to 2013. The data is derived from the Department of Education with over 2000 observations from colleges and universities. The

analysis regarding minority groups poses the hypothesis that graduation rates for minority students will decline throughout universities across the U.S. Based on the data collected it appears that there was an increase in enrollment for Black and Hispanic students, but there was a decrease in graduation rates for Black students. When taking into account other factors such as SAT scores, it can indicate that educational experiences may differ among groups. It could be that differences in graduation rates among minority groups may be the result of limited access to resources that can impact educational success. However, each minority population has varying results when looking at the graduation rates.

Data:

The data that is presented comes from the Opportunity Insights source. The dataset includes the dependent variables as Asian/ Pacific, Black, Hispanic, and Non-resident Alien student populations. It can be said that there is no bias or very minimal bias that is presented from the data considering that it was collected by the Department of Education.

Table 1: Minority Students at Universities

	Descriptive Statistics						
Demographic_Group	**Min**	**X1st_Qu.**	**Median**	**Mean**	**X3rd_Qu.**	**Max**	**NA_s**
Asian/Pacific Share	0	0.00671	0.01497	0.03390	0.03518	0.79159	50
Black Share	0	0.02796	0.06364	0.13037	0.15214	1.00000	50
Hispanic Share	0	0.01165	0.02673	0.07425	0.07482	0.96418	50
Non-resident Alien Share	0	0.00131	0.00987	0.02039	0.02673	0.41468	50

[a] Note: NA_s represents the number of missing values.

In the first table (Table 1) it draws attention to the distribution of Asian/Pacific students among universities. Based on the data these students make up an average of about 3.39% of the student population in some universities. The distribution at some universities indicates that about

13.04% of their population are Black students. The distribution of Hispanic students among universities shows that about on average 7.42% of the population consists of Hispanic students. As for Non-resident Alien students, they represent about 2.04% of the student population. This demonstrates that the average representation of each student population varies within universities. It can also serve as a starting point to address the relationship between minority student populations and graduation rates.

Table 2: Graduation Rate Distribution

	Descriptive Statistics						Graduation Rate	
Demographic_Group	**Min**	**X1st_Qu.**	**Median**	**Mean**	**X3rd_Qu.**	**Max**	**NA_s**	**Grad_Rate**
Asian/Pacific Share	0	0.34485	0.5	0.50085	0.6491	1	1617	0.525
Black Share	0	0.34485	0.5	0.50085	0.6491	1	1617	0.502
Hispanic Share	0	0.34485	0.5	0.50085	0.6491	1	1617	0.488
Non-resident Alien Share	0	0.34485	0.5	0.50085	0.6491	1	1617	0.505

[a] Note: NA_s represents the number of missing values.

Table 1 focused on the distribution of different demographic groups based on the respective proportions in the student body throughout the U.S. Universities. In table 2 it focuses on the Graduation rates of minority groups in order to present the distribution of graduation rates within each demographic group. This table introduces the specific graduation rate (Grad_Rate) for each group and the corresponding “NA_s value” that represents the number of observations used in the analysis. This presents how different minority groups are represented throughout the

universities while also showing how graduation rates vary.

Figure 1: Share of Minority Groups Across United States Universities

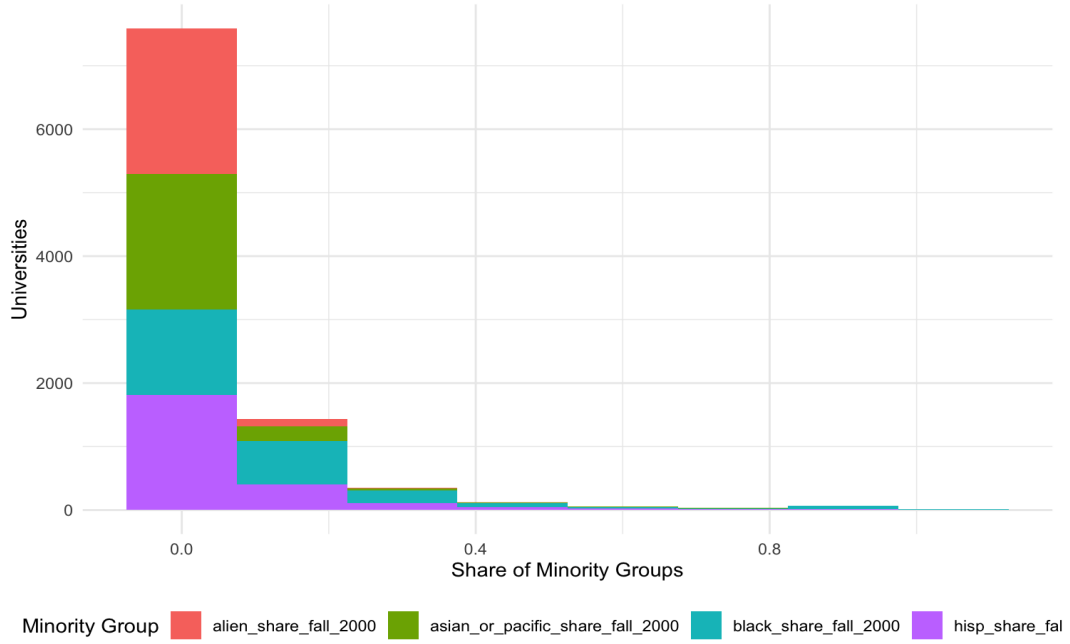

Figure 1 shows a stacked histogram which shows the distribution of the share of different minority groups across universities in the United States. Each bar in the histogram represents the count of universities falling within a specific range of the share of a particular minority group. As the share of minority groups increases the number of universities decreases. There is a small outlier represented at the end of the graph of the black minority group. This is influenced by several historically black colleges in the dataset.

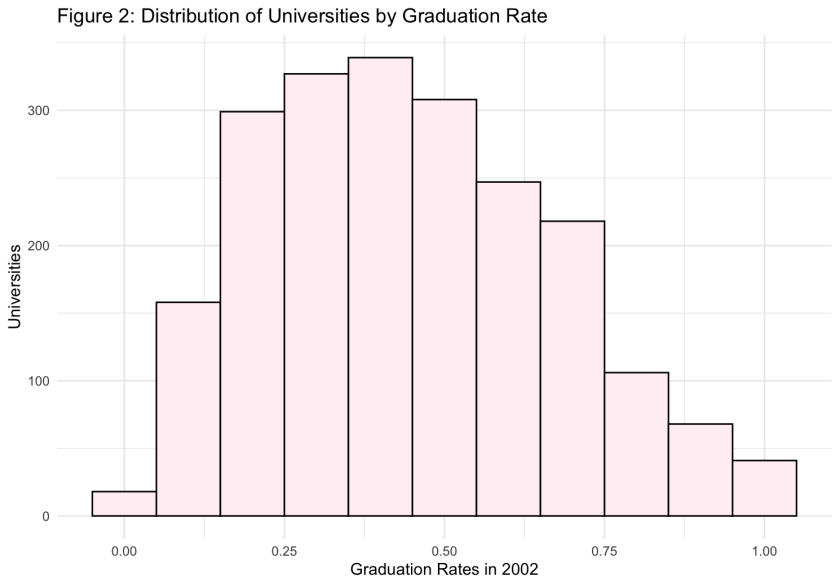

Figure 2 shows a histogram indicating the distribution of graduation rates in 2002 across universities in the United States. A left-skew or negative correlation between the graduation rate and the number of universities demonstrates a disproportionate number of students not graduating from universities. It can be said that more individuals were accepted into the universities than individuals that graduated from said universities.

Methods and Results

Figure 3, 4, 5, and 6 shows the graduation rates of Asian/ Pacific, Black, Hispanic, and Non-resident Alien student populations throughout Universities in the U.S. The scatterplots were meant to represent the relationships between the independent variable (grad_rate_150_p_2002) and the dependent variables. The regression line was added in order to show the trends of the data that was collected. The four scatterplots show the graduation rates from the year 2002 on the x axis and show the percent share of each minority group on the y axis. This is meant to see if there is an impact based on the minority share and graduation rates.

Figure 3 scatter plot presents a positive regression line that suggests that there is a significant correlation between increased graduation rates and Asian/Pacific Islander students. As the population of Asian/Pacific Islander students increases there is also an increase in the graduation rates. This figure does not support the hypothesis since it demonstrates an increase in the graduation rates.

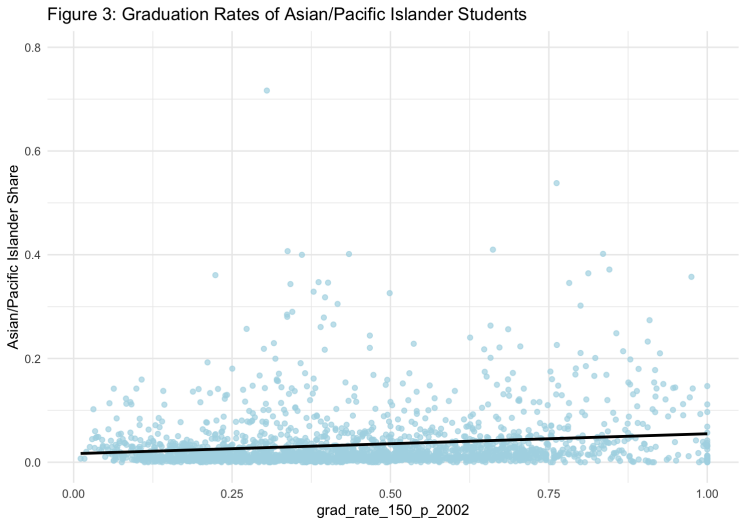

Figure 4 scatter plot shows a negative regression line that suggests that Black students have decreasing graduation rates. It also draws attention toward how the share of Black students can vary based on demographic representation. This scatterplot supports the hypothesis since the regression line shows a decline in Black student graduation rates throughout universities. It may be that there are outside facts that have contributed to the decline in graduation rates for Black students that are not presented in the data.

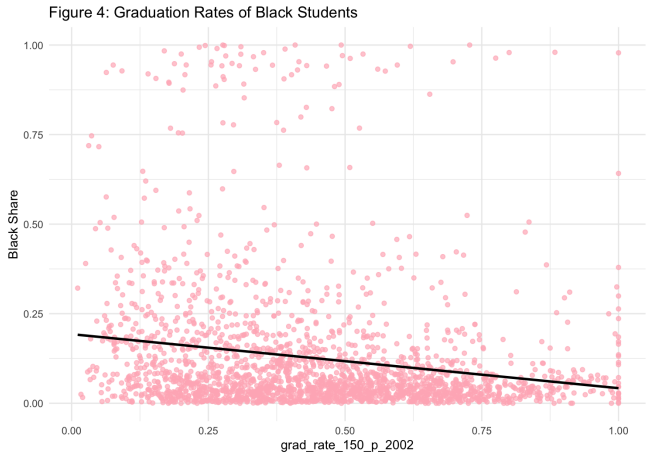

Figure 5 shows the relationship between Hispanic students and graduation rates where the regression line is slightly negative which shows that as there are higher populations of Hispanic students at universities there is also a slight decline in the graduation rates for Hispanic students. The scatterplot supports the hypothesis since the regression line draws attention to declining graduation rates across universities with Hispanic students.

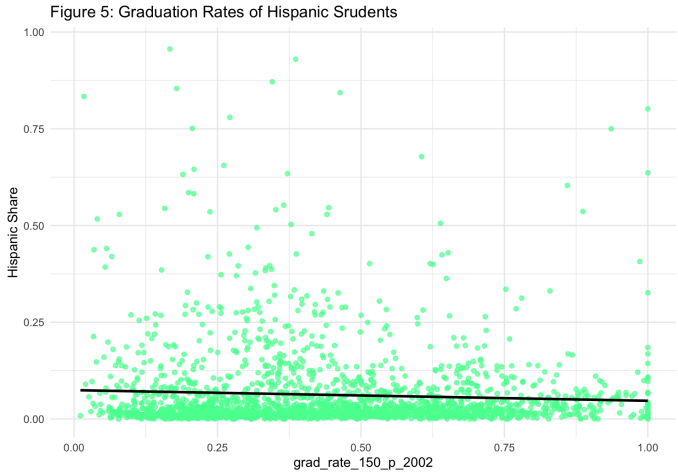

Figure 6 shows Non-resident Alien students present a slight positive correlation between their demographic share and graduation rates. This does not support the hypothesis since the regression line presents a slight increase in the graduation rate. These results draw attention to

educational opportunities considering that the graduation rates represent how each population of minority students shows varying impacts on graduation rates.

Figure 6: Graduation Rates of Non-resident Alien Students

Based on the data presented it may be that Hispanic and Black students may lack educational opportunities which in turn can contribute to the decrease in graduation rates in 2002. In order to better understand the decline in graduation rates, a bivariate regression analysis was used to see the significance of the values.

	Dependent variable:			
	Asian/Pacific Share	Black Share	Hispanic Share	Non-resident Alien Share
	(1)	(2)	(3)	(4)
Enrollment (IPEDS 2000)	0.00000***	0.00000	0.00000***	-0.00000
	(0.00000)	(0.00000)	(0.00000)	(0.00000)
SAT Average 2001	0.0001***	-0.001***	-0.0001***	0.00004***
	(0.00002)	(0.00004)	(0.00002)	(0.00001)
Enrollment (IPEDS 2000)	-0.119***	0.663***	0.128***	-0.014
	(0.018)	(0.045)	(0.021)	(0.009)
Observations	846	846	846	846
R^2	0.173	0.155	0.041	0.026
Adjusted R^2	0.171	0.153	0.039	0.024
Residual Std. Error (df = 843)	0.062	0.157	0.074	0.033
F Statistic (df = 2; 843)	87.956***	77.129***	17.945***	11.431***
Note:				*p<0.1; **p<0.05; ***p<0.01

The regression results presented in the table above presents how the coefficients change based on the share of each minority group. It also shows a one-unit change in the enrollment size. The t-values indicate the significance of each coefficient, and the corresponding p-values

indicate whether the relationship between the enrollment of students and average SAT scores are associated with graduation rates. The coefficient for "SAT Average 2001" for Asian/Pacific Islander students is shown as 0.0001, which is statistically significant at the 0.01 level. This can suggests that a higher SAT average in 2001 is associated with a slight increase in Asian/Pacific Share and graduation rates. The coefficient for "SAT Average 2001" for Black students is shown as -0.001. This is statistically significant at the 0.01 level since it suggests that a higher SAT average in 2001 is associated with a decrease in Black Share and graduation rates. The coefficient for "SAT Average 2001" for Hispanic students is shown as -0.0001. This is statistically significant at the 0.01 level since it suggests that a higher SAT average in 2001 is associated with a slight decrease in Hispanic Share. As for the Non-resident Alien Share/Non-resident students, it shows a coefficient for "SAT Average 2001" at 0.00004. This is statistically significant at the 0.01 level since it suggests that a higher SAT average in 2001 is associated with a slight increase in Non-resident Alien Share. By including average SAT scores in the analysis it represents whether student SAT scores contributed to educational attainment and graduation rate disparities since universities take into account SAT scores when making admission decisions. In some cases students pertaining to minority share may not have the same educational opportunities that other groups of students have.

Discussion

Based on the results shown throughout the analysis, it is evident that there are varying results that occur throughout the populations of minority groups. The hypothesis discussed how graduation rates for minority students will decline throughout universities across the U.S. The hypothesis was correct when looking at the graduation rates for Hispanic and Black students since there was a decline in graduation rates compared to the number of students enrolled. When

looking at the regression model there was also an indication that the average SAT scores for the groups also varied. Though the data only presents evidence of numerical evidence, it is significant to address that there may be other factors that can contribute to the decline in graduation rates for certain minority groups throughout universities in the U.S. That is to say, not all students are given the same educational resources or opportunities to pursue higher education. In many cases, minority students are also associated with low socioeconomic status which may be a factor in the decrease in graduation rates.

When taking into consideration the Asian/Pacific Islander students and the Non-resident Alien students, it can be seen that they had increasing graduation rates among universities in the U.S. a potential reason for this can be due to accessibility to education resources or as a result of supportive academic environments that encourage academic success. These students may be accustomed to receiving educational support. When taking into account SAT scores, it may be that students who had higher average SAT scores in 2000 could have been more academiclly prepared thus showing the relationship with higher graduation rates for these groups. This indicates that by focusing on enhancing standardized test scores and overall academic readiness it could also lead to improved outcomes for Asian/Pacific Islander and Non-resident Alien students.

This may be due to differences in population size, minority dynamics and graduation rates, but not all minority groups are impacted in the same way. By drawing attention to the academic preparedness of certain students it can demonstrate disproportionate outcomes in attaining higher education. The relationships that were shown in the data are significant since they represent the association between the variables, but do not present direct links to each other.

There are outside factors that may contribute to the influence of graduation rates throughout universities.

Conclusion

In conclusion, this report analyzed the relationship between minority students and graduation rates in various universities across the United States, focusing on Asian/Pacific Islander, Black, Hispanic, and Non-resident Alien student populations. The data drew attention to disparities in graduation rates among minority groups. The data showed a decline for Black and Hispanic students, while Asian/Pacific Islander and Non-resident Alien students showed increasing graduation rates. These findings for Black and Hispanic students supported the hypothesis that graduation rates for minority students would decline throughout U.S. universities, suggesting the presence of systemic factors impacting educational experiences for different minority groups. However, in the case of Asian/Pacific Islander students and Non-resident Alien students the hypothesis was rejected since they showed an increase in graduation rates.

The analysis also highlighted the potential influence of factors such as enrollment size and average SAT scores on graduation rates. Based on the potential outside factors, there may be indications of unequal access to resources and academic support that can contribute to disparities among some minority groups. Low socioeconomic status among some minority students may also play a role in the decline in graduation rates. On the other hand, the higher graduation rates among Asian/Pacific Islander and Non-resident Alien students suggests the existence of supportive academic environments or better access to education resources for these groups. Improving standardized test scores and academic preparedness might lead to improved outcomes for these students.

It is important to acknowledge that the relationships identified in the data do not represent direct causal links, as there may be other external factors that can also influence graduation rates throughout universities. The varying impacts on different minority groups emphasize the need to address the educational disparities faced by minority students and work towards creating more equitable opportunities and resources in higher education.

References

Opportunity Insights. "College Level Characteristics from the IPEDS Database and the College Scorecard." Opportunity Insights, 2013. https://opportunityinsights.org/paper/mobilityreportcards/